PY-STEM · ACTIVITY PYS-11 · COMPUTING

BookBot Bin Logic Challenge

Retrieve the right book by using bin addresses, not a conveyor race.

Win condition: Most correct retrievals with few traffic conflicts and clear algorithm defense.

THE SCIENCE YOU ARE USING

Automated storage and retrieval. Temple's Charles Library BookBot stores books in bins identified by an address, not by subject on a shelf. A crane fetches the bin by its address on request. The same logic runs warehouses: store anywhere, remember the address, retrieve on demand.

Routing and search. Good addressing and smart routing minimize travel and avoid traffic jams. You simulate storage and retrieval with order cards, bin addresses, and routing rules, then defend your algorithm.

HOW TO BUILD AND RUN IT

01 **Set up the bins.** Letter the rows down the side (A-D) and number the columns across the top (1-6) so an address like B3 is clear. Place items by address, not by subject.

02 **Read an order.** Draw an order card listing the items to retrieve and their addresses.

03 **Plan the route.** Order your retrievals to minimize travel and avoid collisions with other teams.

04 **Run the retrieval.** Fetch the items by address against the clock. Wrong items cost points.

05 **Defend the algorithm.** Explain the rule you used to choose the order and why it is efficient.

Safety: Keep clear of actual library operations. Tabletop version preferred after tour. Allergy: None.

MATERIALS

Bin cards	set
Address labels	shared
Order deck	per team
Route mat	per team
Timers	per team
Grabber tool	optional

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________________

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The ONE variable we are testing: ______________________________

Baseline result: ______________________________

After redesign: ______________________________

DEFEND YOUR DESIGN

What evidence made you change your design? ______________________________

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If you ran it again, what is the ONE thing you would change? ______________________________

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HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Low traffic conflicts	25
Routing or algorithm explanation	20
Efficiency	10
Teamwork	10
Total	**100**